

Willian Vieira PhD

PhD quantitative ecologist with data engineer experience. I build statistical models, reproducible workflows, and environmental data infrastructure that turn complex biodiversity and geospatial data into analyses people can inspect, trust, and reuse.

DATA SCIENCE & ENGINEER EXPERIENCE

- 2025
|
2024
(1 yr 9 m)

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Data Analyst
Habitat, Montreal, Canada
Delivered statistical and ML models, spatial analysis, and reproducible reports for environmental consulting projects using client and open geospatial data | Became the de facto lead for analytical infrastructure, standardizing R/Python/Julia packages, documentation, code review, Docker/Unix environments, and CI/CD automation | Moved heavy spatial workflows toward cloud-backed access patterns, using object storage and range-request-friendly data access to reduce fragile network-drive dependency | Designed metadata and pipeline conventions to improve lineage, validation, debugging, handoff, and reuse across recurring analyses
- 2024
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2017

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PhD Research
Integrative Ecology Lab, Sherbrooke, Canada
Developed Bayesian hierarchical and nonlinear statistical models in R and Stan to represent heterogeneous biological populations, forecast dynamics, and propagate uncertainty from noisy, biased, incomplete real-world data | Authored open-source R packages (🔗, 🔗) for demographic modelling, with version control, automated tests, documentation, and reusable APIs | Published a **technical methods book** walking from mathematical concept to R implementation to model evaluation, alongside one peer-reviewed **publication** and two preprints (📄, 📄) | Ran thousands of simulations on HPC infrastructure (🔗) to test model behavior, compare assumptions, and validate inference at scale | Built reproducible R pipelines harmonizing climate, remote sensing, and field-inventory data across North America
- 2022
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2020
(part-time)

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Biostatistician
Environment and Climate Change Canada - Quebec, Canada
Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **data pipeline** with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

EDUCATION

- 2024
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2017

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PhD, Ecology
Université de Sherbrooke - Sherbrooke, Canada
📖 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*
- 2016
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2015

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Masters 2, Agroecology and Resource Management
Bordeaux Sciences Agro, Bordeaux, France
📖 *Modelling the dispersion of weed species in agricultural landscapes*
- 2015
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2010

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BSc in Agronomy
Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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✉ w.vieiraw@gmail.com
in [Linkedin/WillVieira](#)
🔗 [GitHub/WillVieira](#)
📄 [Publications](#)

SKILLS

Statistics, Modelling & Ecology:

Bayesian statistics · Nonlinear hierarchical models · SDMs · Sampling design · Spatial bias · Simulation · Sensitivity analysis · Uncertainty quantification · Model evaluation

Environmental & Geospatial Data:

Biodiversity monitoring · Forest inventory data · Climate and remote-sensing covariates · Habitat and vegetation data · Environmental assessment support · Decision-ready maps and reports

Programming & Scientific Software:

R · Python · Stan · SQL · Bash · DuckDB · Julia · Package development · Shiny · Quarto

Data Engineering & Reproducibility:

ETL pipelines · Metadata and data governance · Cloud object storage · Cloud-optimized geospatial access · Automated validation · Docker · CI/CD · Git · Unix · HPC · Make · Reproducibility

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional



TEACHING EXPERIENCE

- 2021**
 - **Biodiversity Modelling Summer School**
Bios² summer school  Sherbrooke, CA
Teaching assistant and lectured for graduate students Evaluating ecological models using Bayesian statistics
- 2020**
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2019
 - **Methods in computational ecology**
BIO500 - Département de biologie, Université de Sherbrooke  Sherbrooke, CA
Teaching assistant and lectured for undergraduate students Tools for reproducible science | SQL | Git | R | LaTeX | Markdown | Makefile
- 2020**
 - **Software and Data Carpentry workshop**
Bios² training for graduate students  Online
The Unix Shell Data Management with SQL for Ecologists
- 2019**
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2018
 - **QCBS R Workshops**
Université de Sherbrooke & Université du Québec à Rimouski  Sherbrooke & Rimouski, CA
Instructor to graduate students **Workshops:** Introduction to R | Data management | Data Visualization | Control flow | Linear models
- 2019**
 - **Probability and statistics**
GCI145 - Faculté de génie, Université de Sherbrooke  Sherbrooke, CA
Teaching assistant and lectured for undergraduate students Probability theory | Distribution | Descriptive statistics | Hypothesis testing | Linear models
- 2018**
 - **Modelisation de la dynamique forestier du Quebec face au changement climatique**
Département de biologie, Université de Sherbrooke  Sherbrooke, CA
Guest lecturer in Plant Ecology (**Presentation**) to undergraduate students Covered modelling approaches in ecology | model construction and applications
- 2018**
 - **Introduction to scientific programming**
BIO109 - Département de biologie, Université de Sherbrooke  Sherbrooke, CA
Teaching assistant for undergraduate students Pogramming concepts | Data management | Control flow | R language



TRAINING

- 2022**
 - **Data visualization and biodiversity modelling**
Summer school - Bios²  Magog, CA
Business intelligence | data user profiles | Conceptualize interactive dashboard | Figma
- 2020**
 - **Mathematical Modelling in Ecology and Evolution**
Workshop - Bios²  Remote
Developing equations | Finding equilibria | analyzing stability | Maxima
- 2019**
 - **Stage-based demographic models in ecology, evolution and conservation biology**
NERC Advanced Training - University of Oxford

Theory and practice | Matrix population models | Integral Projection Models

- 2019 ● **Biodiversity Modelling**
Summer school - Université de Sherbrooke 📍 Magog, CA
 Theory and practice on different biodiversity modelling techniques Statistics | Differential equations |
 Stochastic simulations | Climate change models | uncertainty tracking
- 2017 ● **Data-driven ecological synthesis**
DDES summer school - Université de Montréal 📍 Montréal, CA
 Good practices in scientific computing Data management | Mathematical tools | Data analysis tools |
 Parallel computing | Reproducible science
- 2017 ● **Bayesian Statistics for Ecologists**
Summer school - Université de Sherbrooke 📍 Magog, CA
 Theory and practice using Stan | Probability theory | Likelihood | MCMC | Hierarchical modelling | Model
 comparison



PUBLICATIONS

- 2026 ● **Unifying individual and metapopulation scales with stochastic population models: the effect of climate and competition on tree range limits**
 In production. Preprint hosted on
[Vieira, W.](#), Gravel, D.
- 2026 ● **Sensitivity of tree species demography to climate and competition across their range**
 Under review. Preprint hosted on
[Vieira, W.](#), MacDonald, A., Gravel, D.
- 2024 ● **Paying colonization credit with forest management could accelerate the range shift of temperate trees under climate change**
 Ecological Modelling
 • [Vieira, W.](#), Boulangeat, I., Brice, M., Bradley, R.L., Gravel, D.
- 2020 ● **Moderate disturbances accelerate forest transition dynamics under climate change in the temperate–boreal ecotone of eastern North America**
 Global Change Biology
 • Brice, M.H., Vissault, S., [Vieira, W.](#), Gravel, D., Legendre, P., Fortin, M.J.
- 2020 ● **Genetic and demographic aspects of *Varronia curassavica* Jacq. in a heterogeneous coastal ecosystem**
 Anais da Academia Brasileira de Ciências
 • Hoeltgebaum, M.P., Lauterjung, M.B., Montagna, T., Candido-ribeiro, R., [Vieira, W.](#), Bernardi, A.P., Cristofolini, C., Reis, M.S.D.